

Modeling Pyroclastic Density Currents

- Pyroclastic density currents (PDCs) are among the most deadly volcanic phenomena.

- Mitigation of risk associated with PDCs depends upon accurate forecasting of possible inundation areas, often using mobility metrics, the distribution of past deposits, or flow models.



What are PDCs?

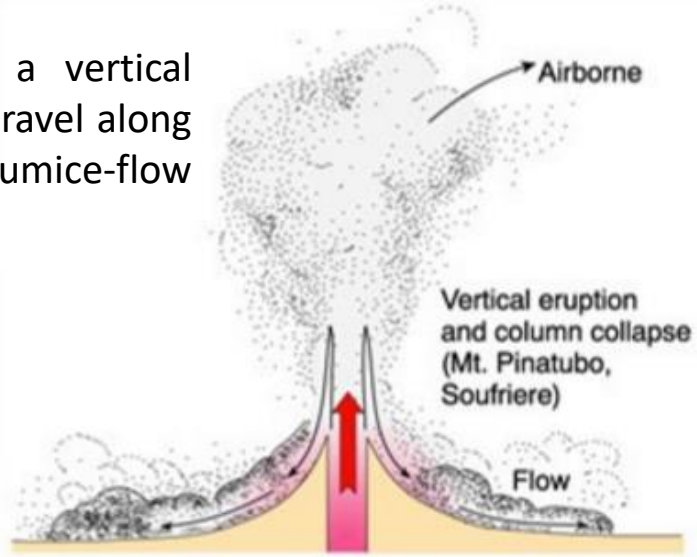
Pyroclastic Density Currents (PDCs)

- Fast-moving, hot, ground-hugging mixtures of gas and particles (up to v. large blocks)
- Extremely destructive and cause lots of deaths
- Includes pyroclastic flows, surges, blasts, ash flows, nuée ardentes, block-and-ash flows, pumice flows, etc. Often, the deposits have different names than the flows (a wide range of names are used interchangeably and confusingly). PDC is a catch all term that I'll use.





Column-collapse PDCs are formed by the collapse of a vertical explosive column that falls back to earth and continues to travel along the ground. They form pumice and ash rich ignimbrite or pumice-flow deposits, that are radially distributed around the volcano.





Mayon, Philippines



Lascar, Chile

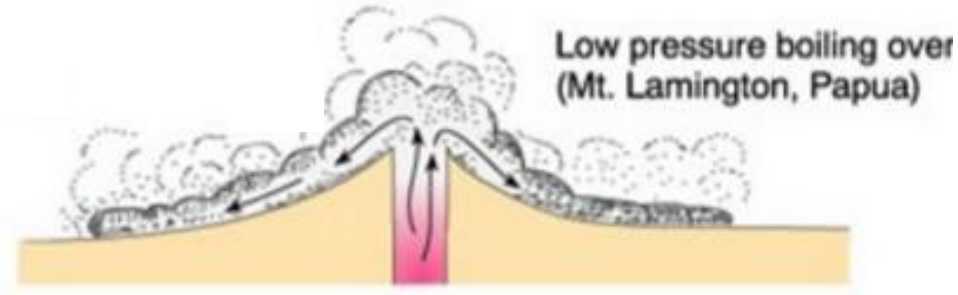


Pinatubo, Philippines

Sometimes, pressures are not great enough to form a full vertical column and result in “boiling-over” style activity, which produces PDCs similar to those from column-collapses.



Tungurahua, Ecuador



Fuego, Guatemala













Instead of being directed vertically, blasts can be directed horizontally and form strongly erosive lateral blast PDCs.



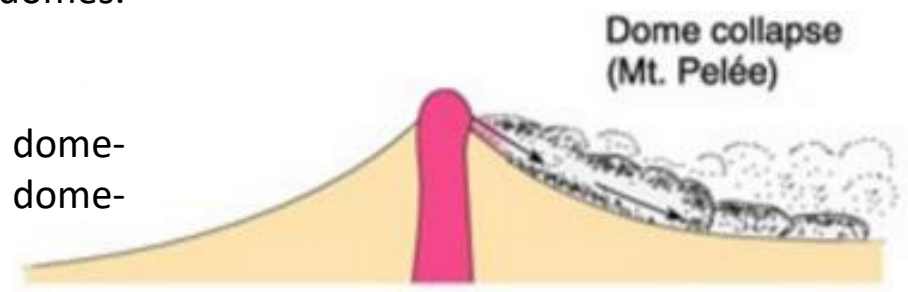
Soufrière Hills Volcano, Montserrat

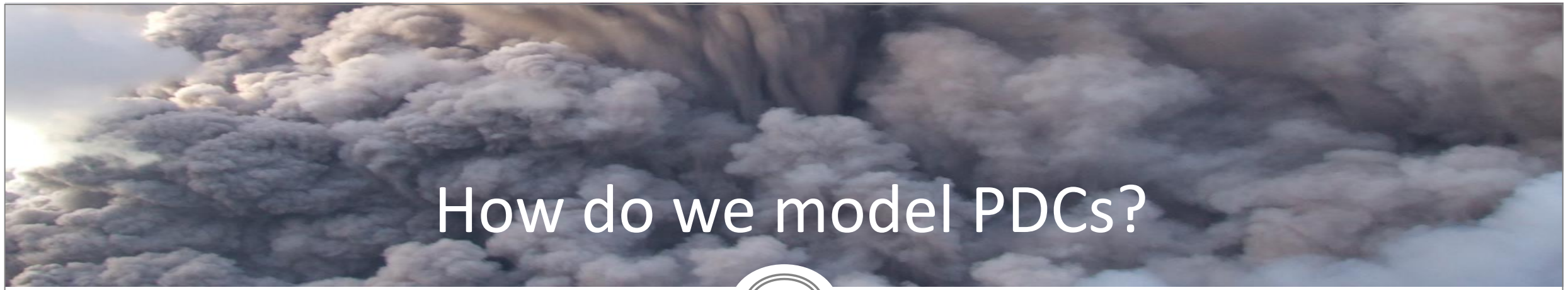


Sinabung, Indonesia

High-particle concentration PDCs are generated by the gravitational or explosive collapse of lava domes. These flows form block-and-ash flow deposits.

Sometimes see “Pelée type” (explosive dome-collapse) or “Merapi type” (gravitational dome-collapse)

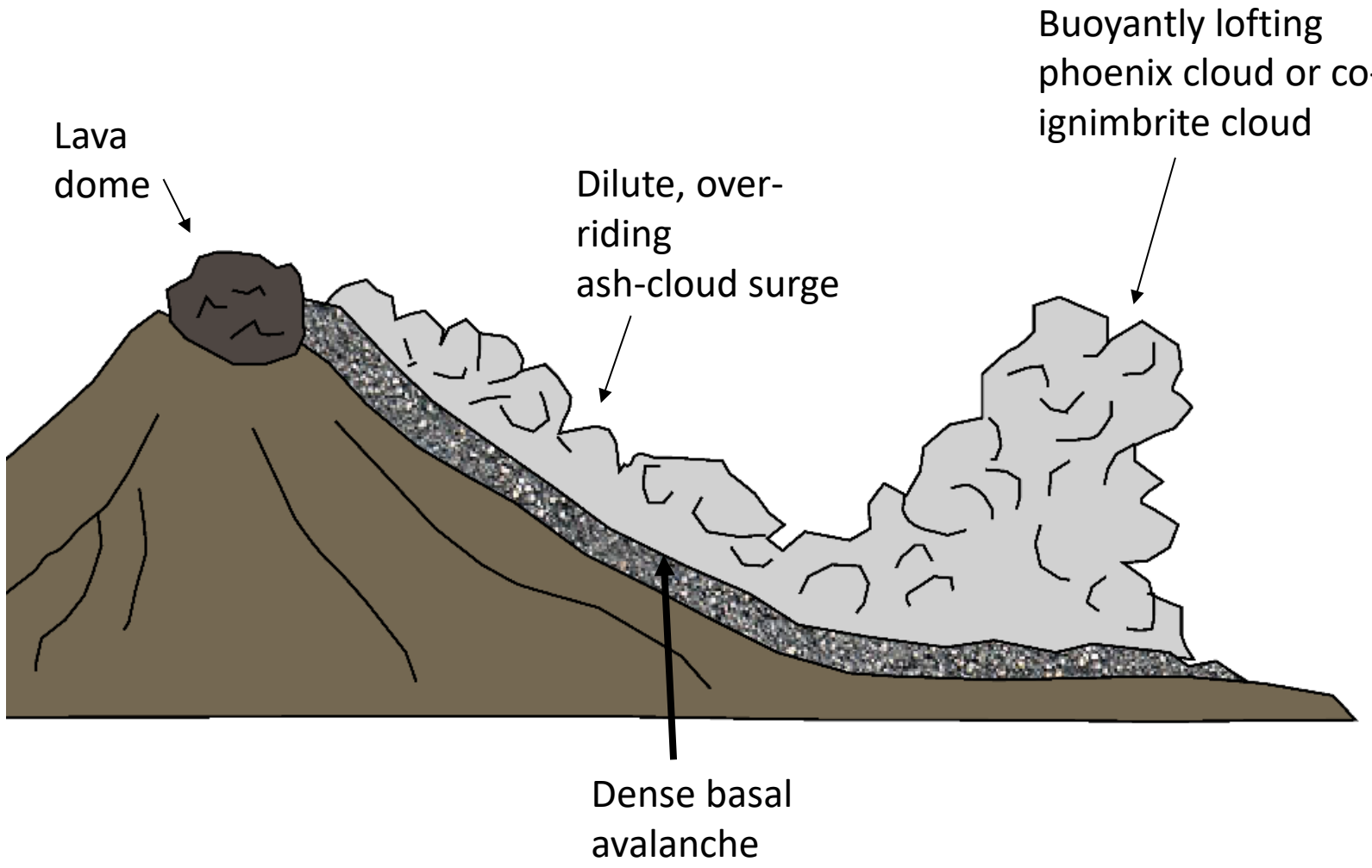




How do we model PDCs?

Parts of a PDC

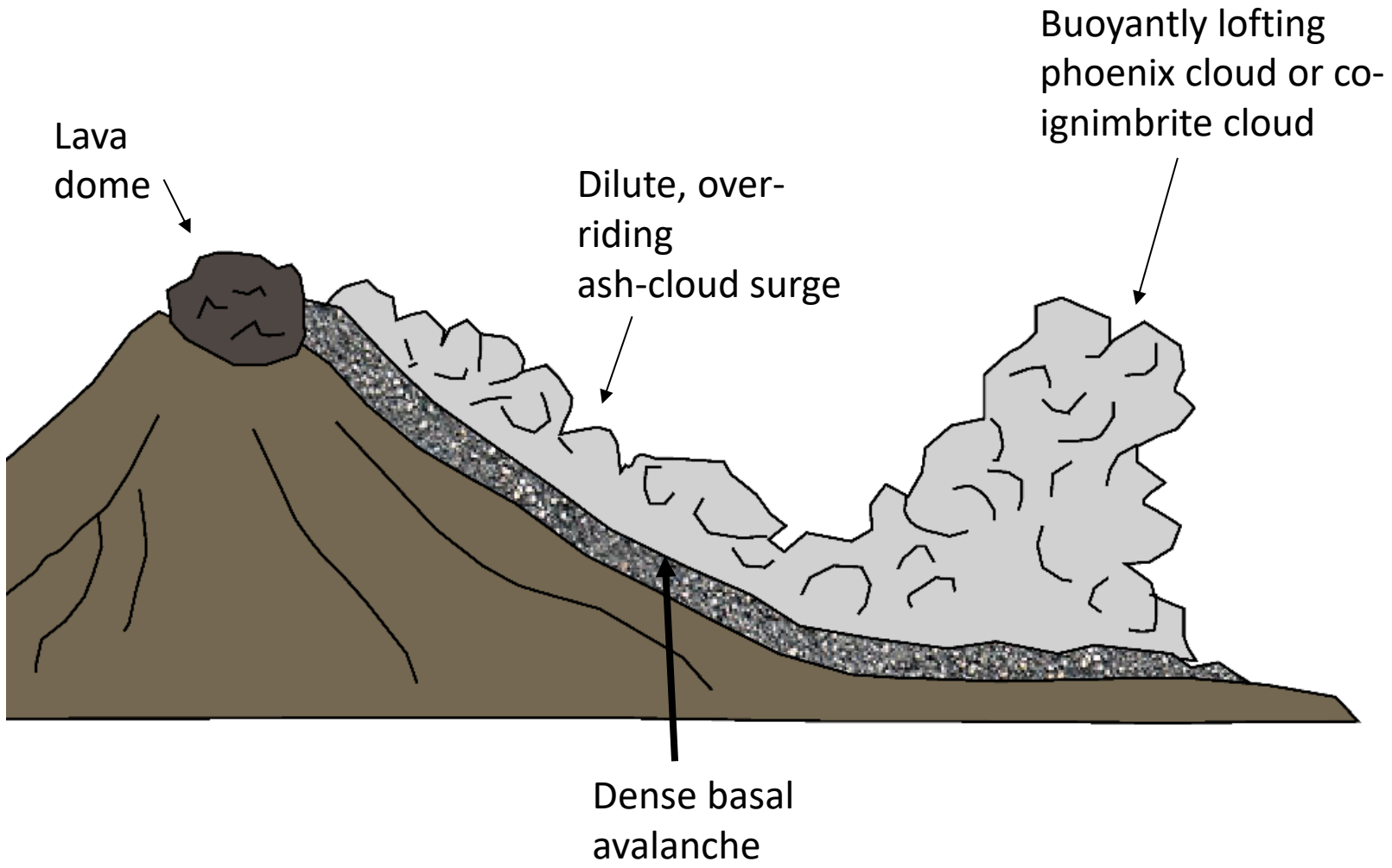
PDCs are understood to be density-stratified currents which, to varying degrees, develop high concentration, dense undercurrents (dense basal avalanches) and dilute, turbulent, overriding ash clouds (ash-cloud surges).



Parts of a PDC

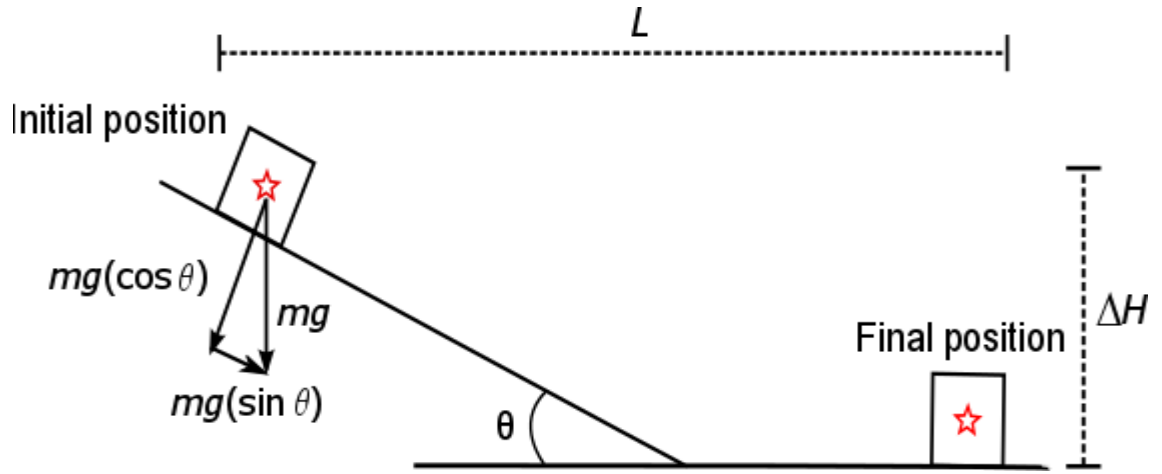
Dense basal avalanche usually follows along valleys (but sometimes escapes)

Ash-cloud surge often spread outside of channels, and sometimes can detach from the dense basal part and go in a different direction



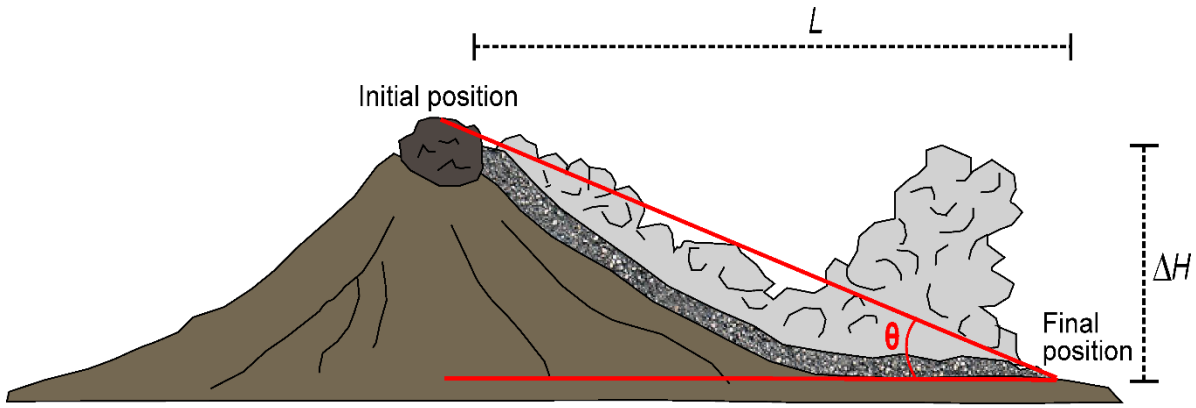
ΔH/L or Heim coefficient

- Height dropped/runout length
- Block begins to slide when gravitational force down a plane equals the frictional force
- Tanθ =coeff. of friction, μ = ΔH/L = Heim coefficient



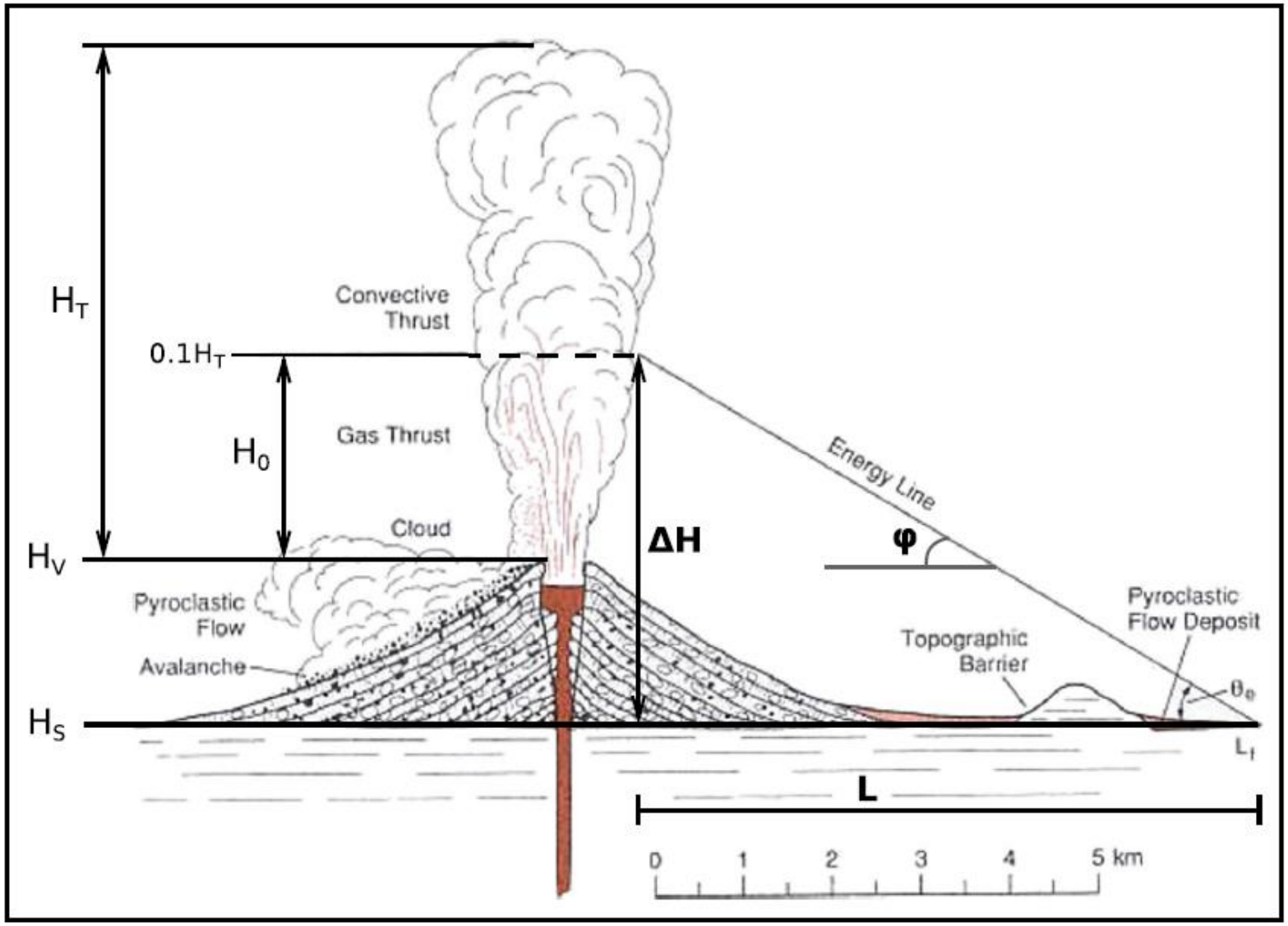
Block begins to slide when mgsinθ = μmgcosθ

Coeff. of friction, μ = $\frac{mgsin\theta}{mgcos\theta}$ = tanθ



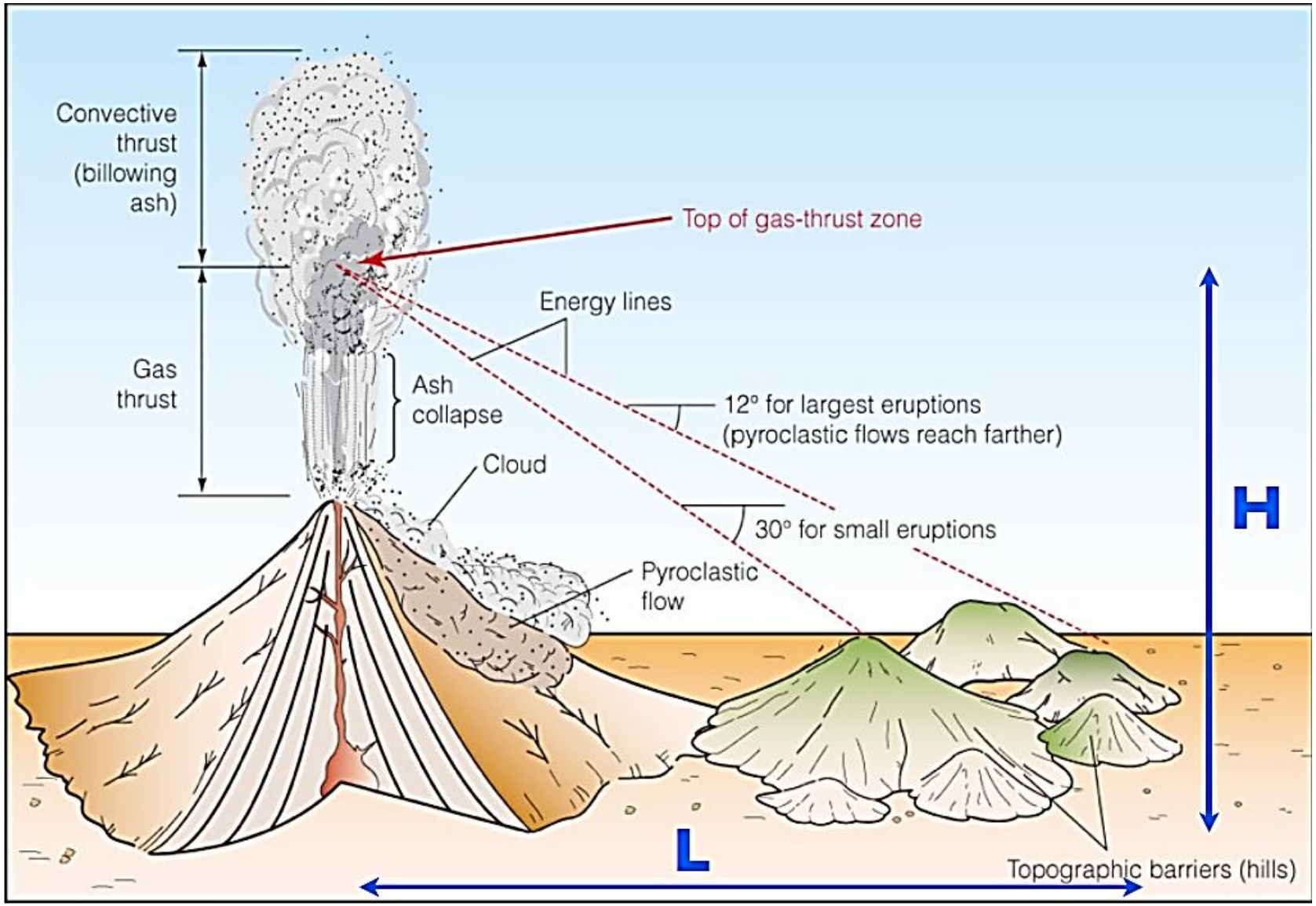
ΔH/L or Heim coefficient

- For column-collapse flows, the H doesn't start at the top of the dome, it starts from where the collapse happens, at the top of the 'gas thrust' region
- Column-collapse PDCs usually collapse 500-2000 m above the summit



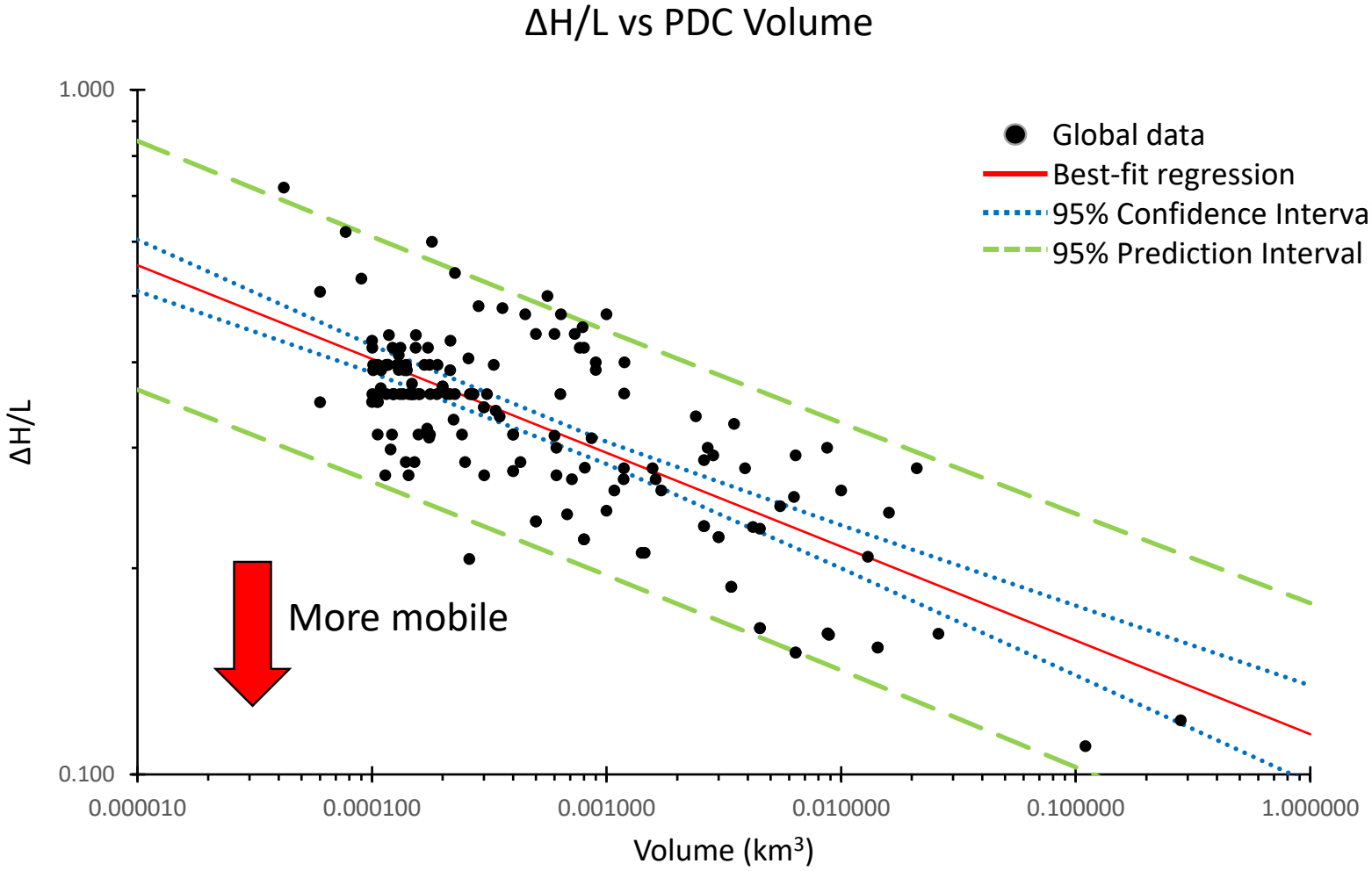
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ΔH/L or Heim coefficient

- Should be independent of mass/volume, and depend on material propertied but actually ΔH/L shows an inverse relationship with volume
- Large volume flows are more mobile than small volume flows
- Lots of different theories to explain this 'excess mobility'. Large volume flows have complex rheologies that are difficult to simplify in any model.
- Lots of scatter in the data

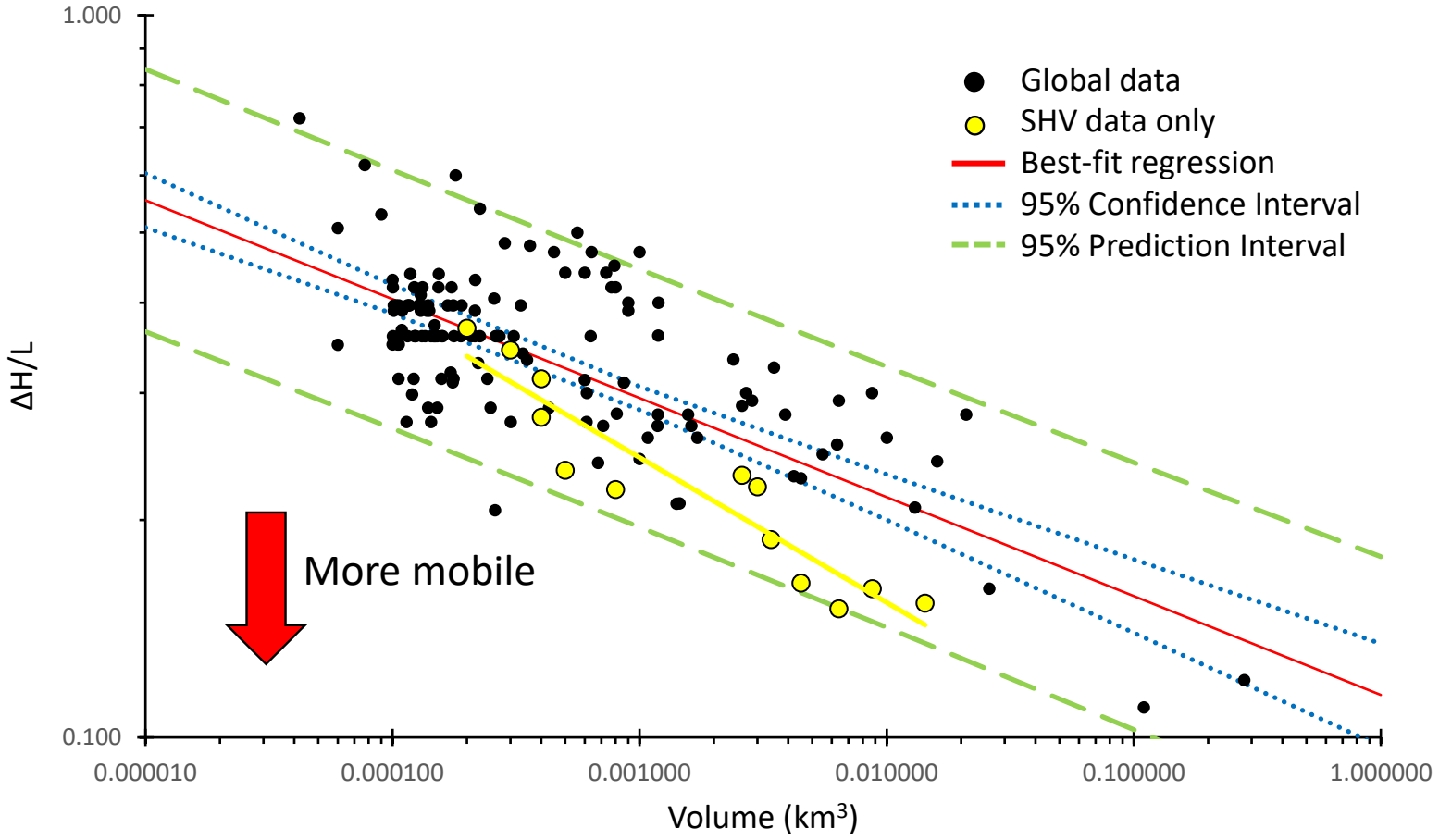


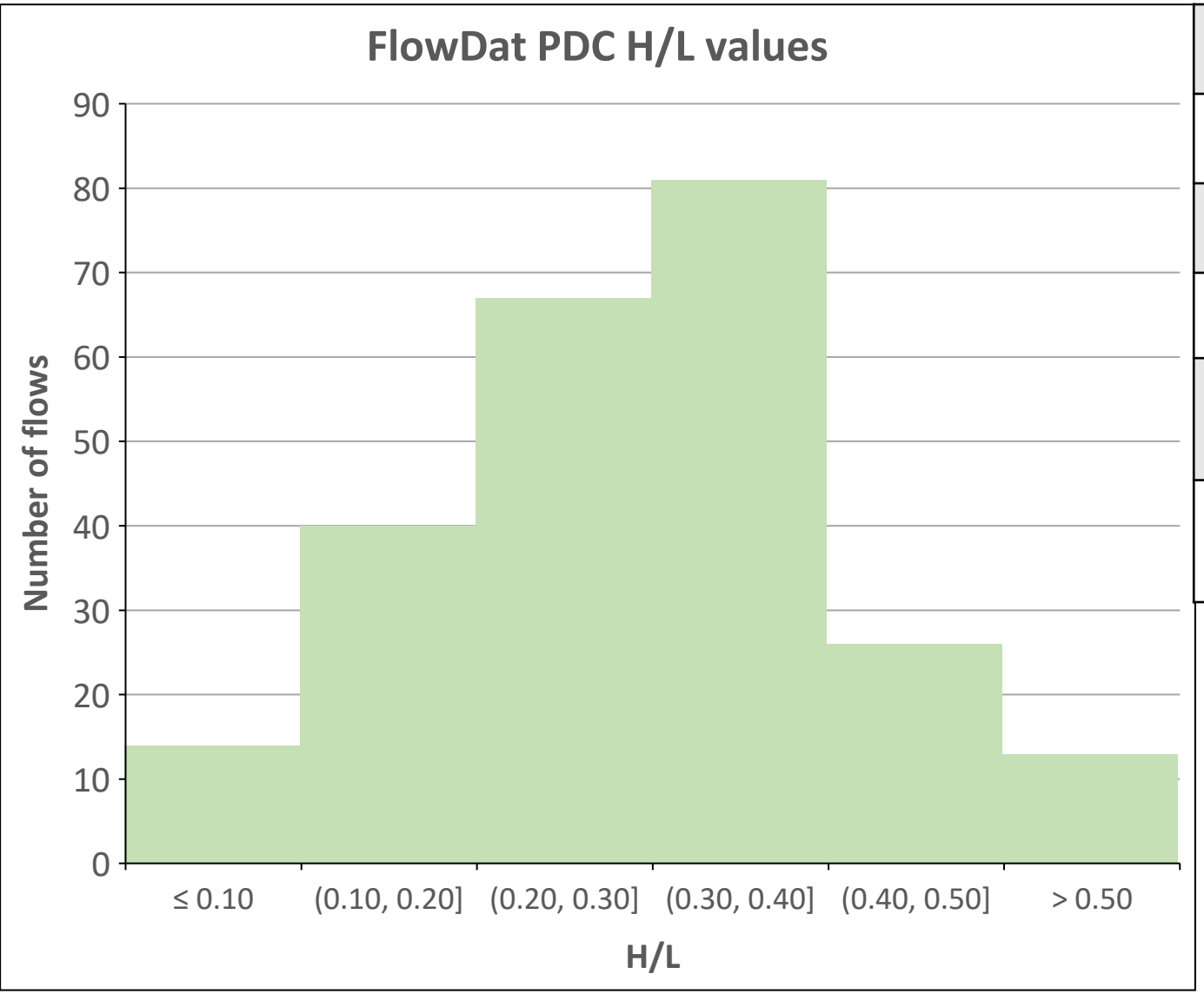
Best fit: $\Delta H/L = 0.10(Volume^{-0.15})$

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ΔH/L or Heim coefficient

- Different volcanoes can have differences in mobility
- For example, PDCs at SHV in Montserrat are more mobile than elsewhere, probably because of channelization and other topographic effects





#	241 (with H/L values)
Avg.	0.30
Median	0.30
Min.	0.04 (very large ignimbrites)
Max.	0.80 (very small dome-collapses/rockfalls)
Middle 80%	0.15-0.44

Most PDCs: 0.2-0.4

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ΔH/L ranges for dome- and column-collapse PDC energy cones from FlowDat regressions

Volume (km^3)	Volume (10^6 x m^3)	ΔH/L best fit Dome collapse	ΔH/L best fit Column collapse
0.0001	0.1	0.39	0.46
0.001	1	0.28	0.43
0.01	10	0.21	0.27
0.1	100	0.15	0.17

Dome-collapse PDCs (based on 123 flows): $\Delta H/L = 0.108\left(V^{-0.140}\right)$

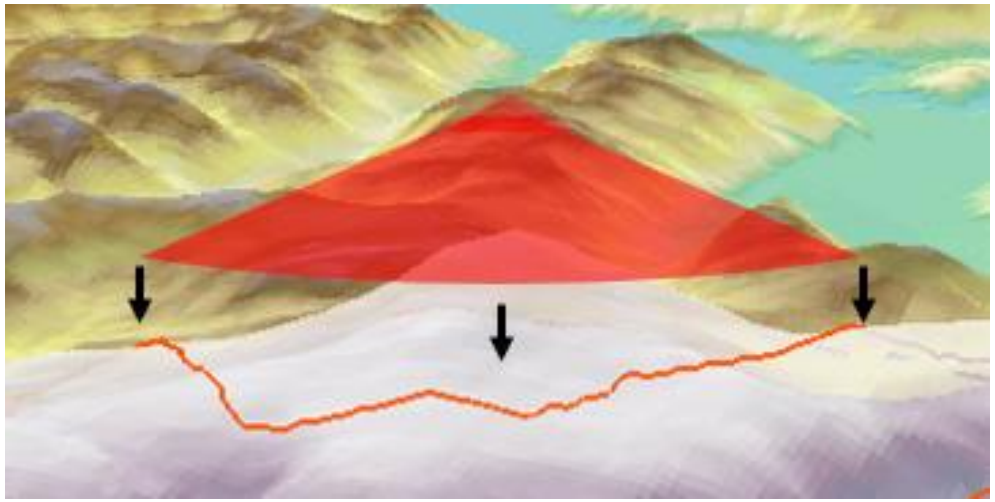
Column-collapse PDCs (based on 16 flows): $\Delta H/L = 0.106\left(V^{-0.201}\right)$

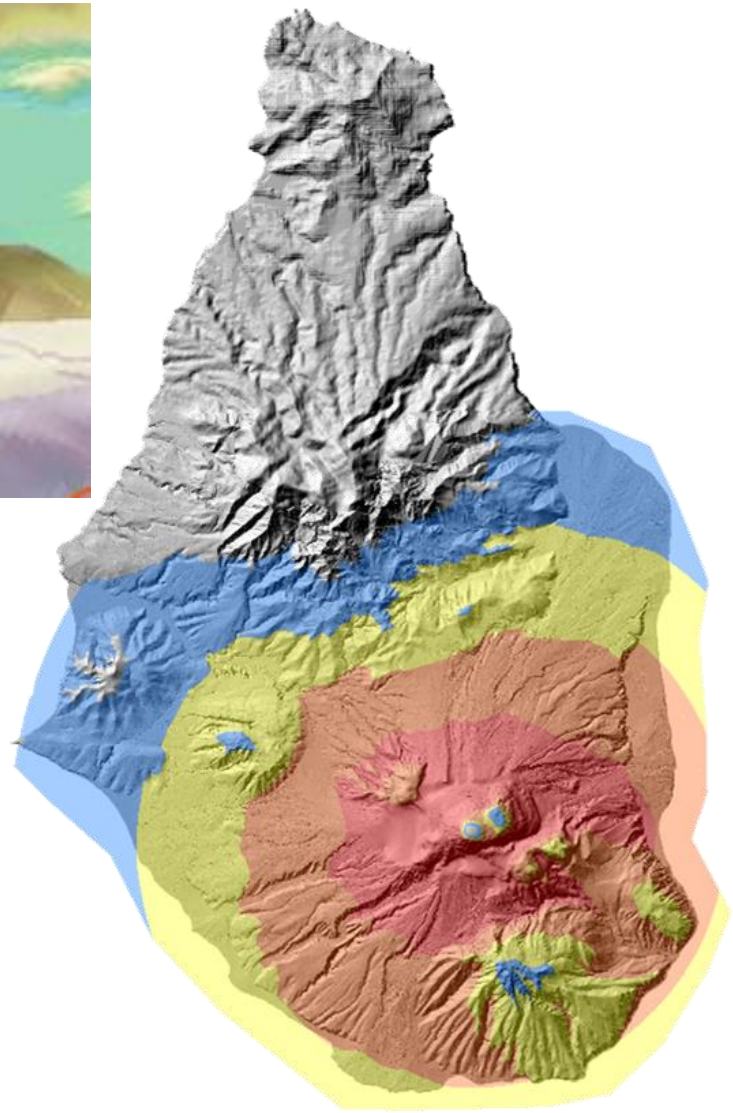
All PDCs (based on 156 flows): $\Delta H/L = 0.108\left(V^{-0.141}\right)$

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Energy Cone

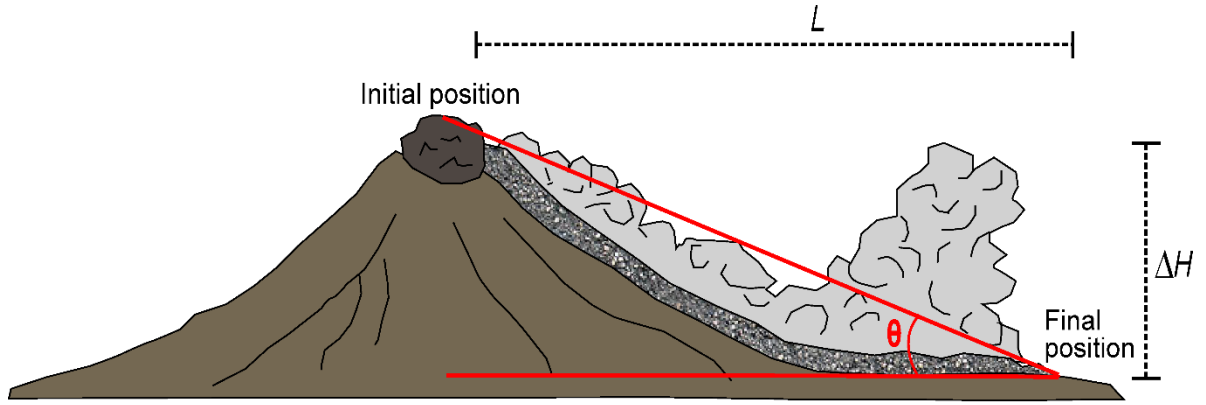
- Also called ΔH/L cone
- One of the first modeling techniques applied to PDCs
- Uses the ΔH/L vs. volume relationship to draw a line from the summit of the volcano to the estimated extent of PDCs using the ΔH/L ratio
- Can be done by hand on a map, or using a GIS program
- Used many places, often in combination with other modeling techniques

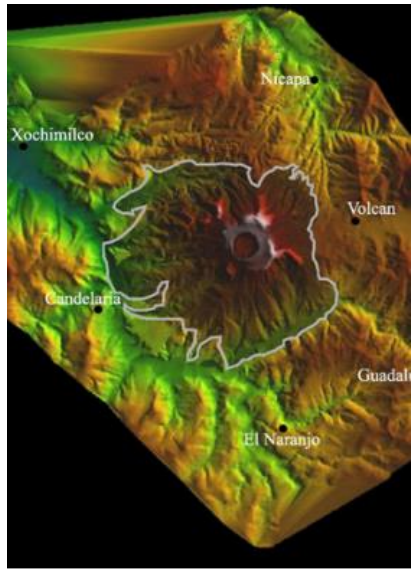




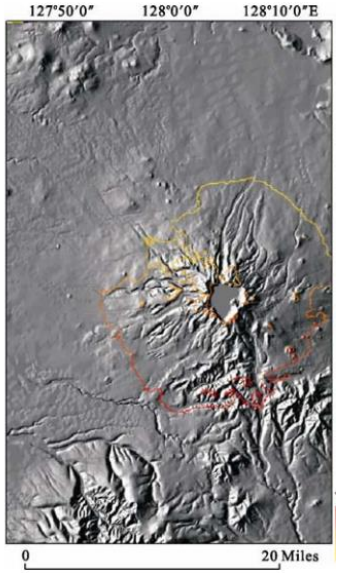
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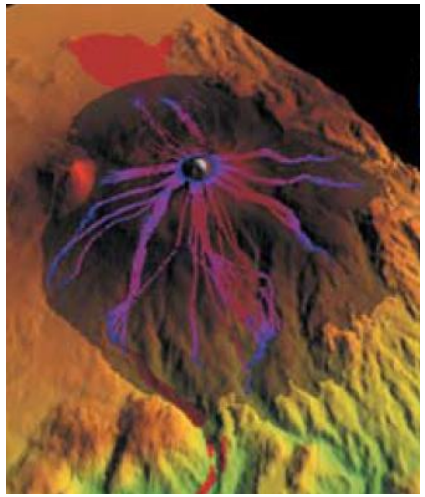




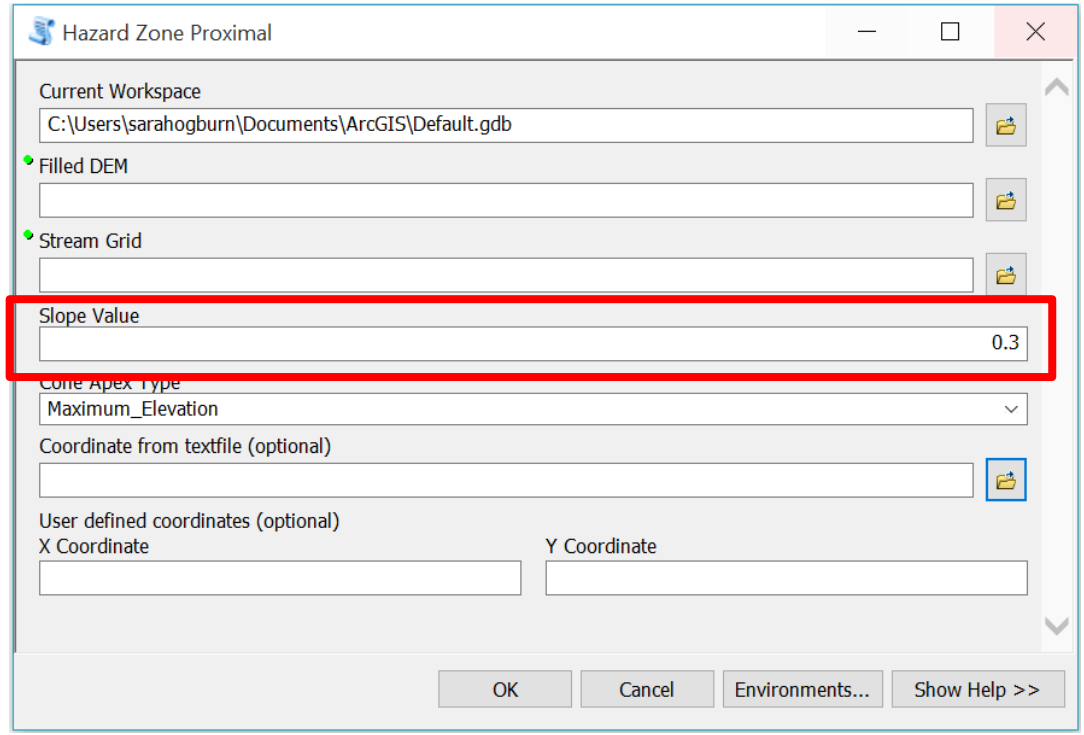
Energy cone at El Chichón, Mexico
Macías et al. 2008

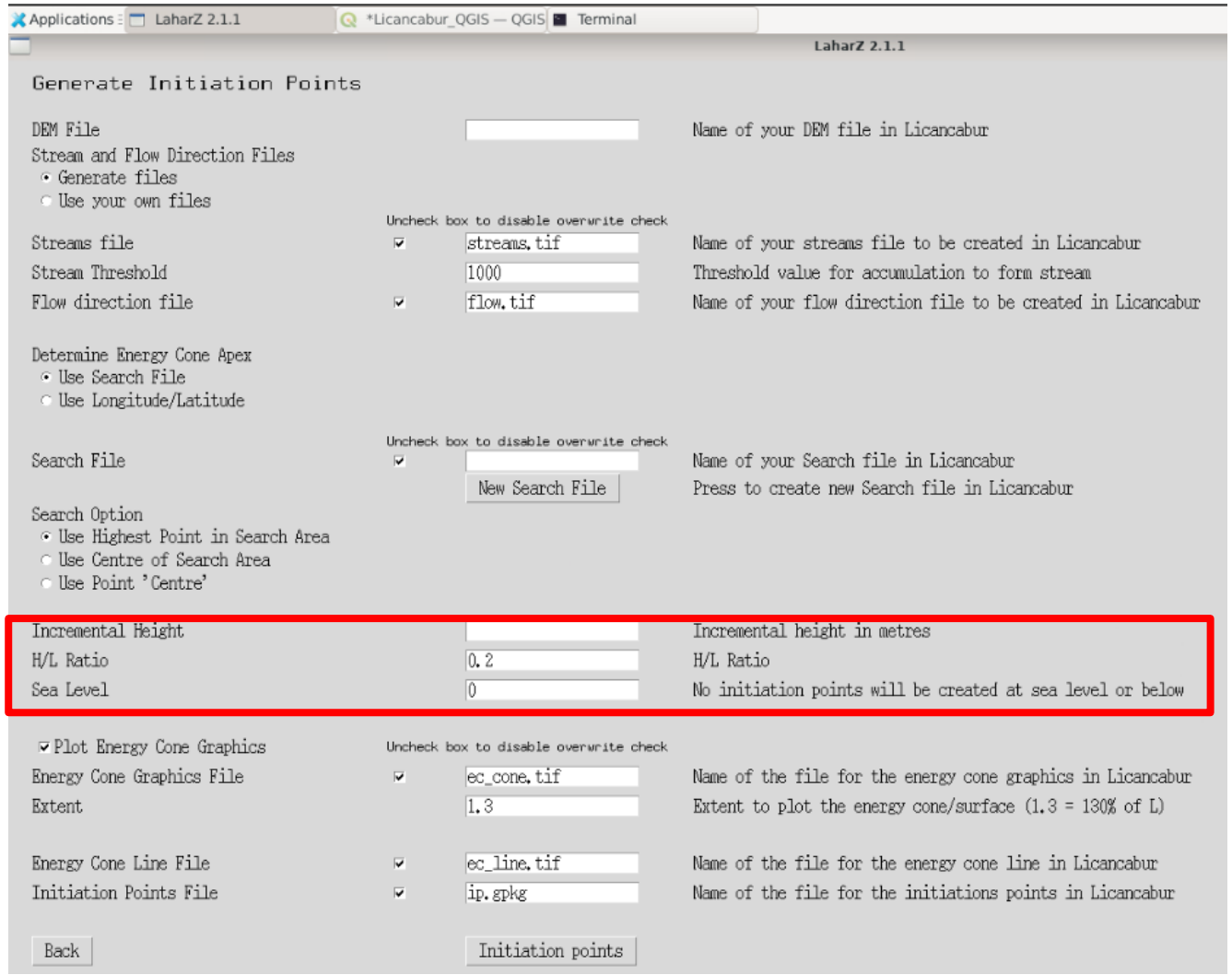


Energy cone at Tianchi volcano, China
Wan et al. 2012



Energy cone & FLOW2D at Citlaltépetl, Mexico
Sheridan & Macias, 2004



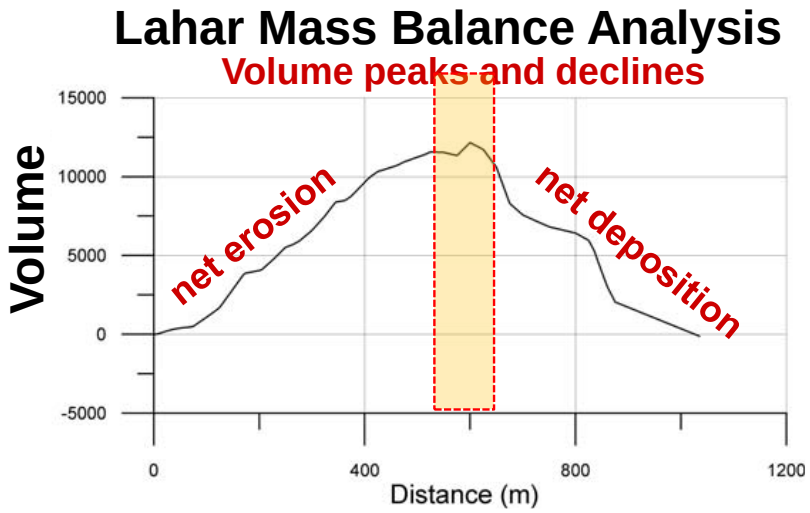


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Proximal Hazard Zone for LAHARZ

Why use H/L Cone for lahars?

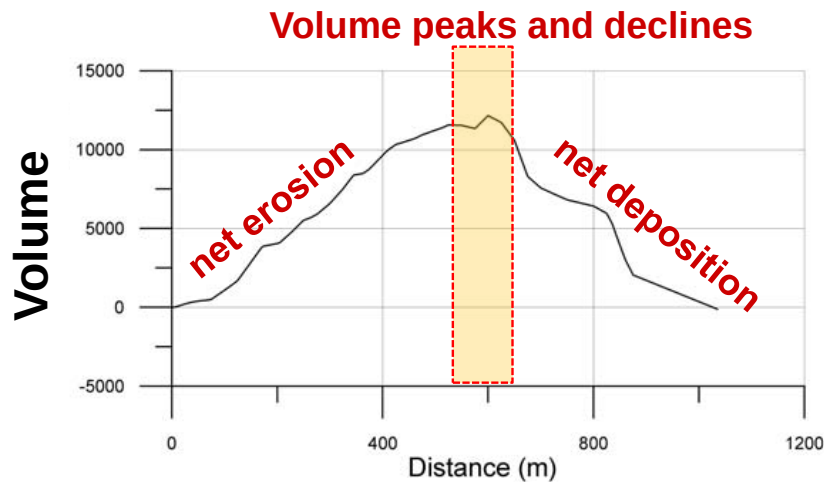
- Lahars may initiate anywhere on the volcano edifice. We can't know an exact start location in advance.
- Loose pyroclastic flow deposits and ashfall are often the source material for lahars and are most abundant within the H/L cone.
- Lahars gain volume (entrain material) on steep slopes.



- So, we start modeling lahars by estimating where on the flanks the flow volume is greatest.
- We do this by defining an H/L cone that marks the transition between erosion and deposition. That is, inside the H/L cone, the volume is still increasing. Outside the H/L cone, the volume is maintained or decreasing (depositing).
- How do we identify an H/L cone that represents this transition?
    - from net erosion to net deposition
    - from steep slopes to moderate slopes (20°-30°), that is, find the break in slope
    - from pyroclastic debris fans to more channelized flows

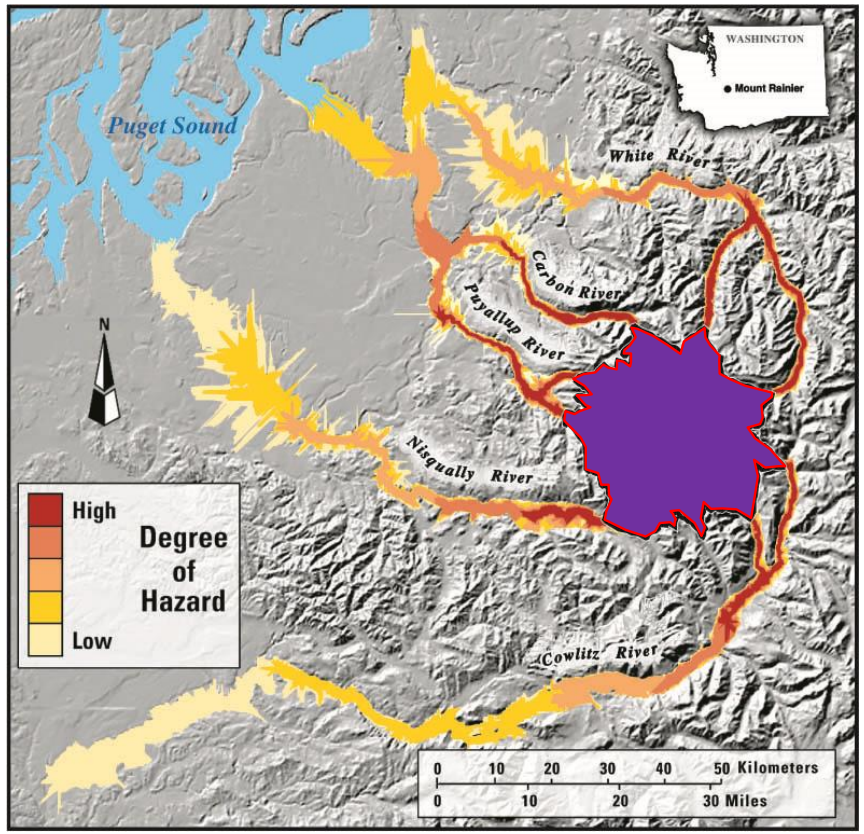
ΔH/L cone for LAHARZ proximal hazard zone

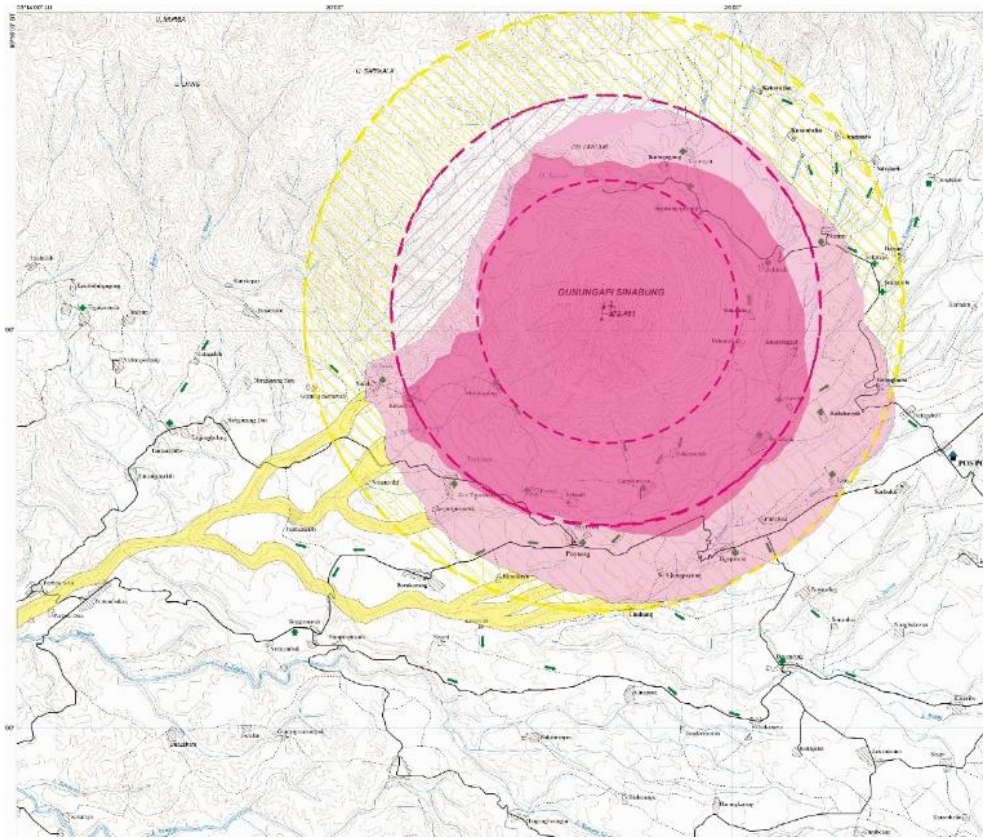
- Trying to find the ΔH/L cone where erosion ends and deposition begins (break in slope) in order to start the LAHARZ model
- Has to do with the shape of the volcano, where most PDCs inundate, where the slopes are steepest
- Depending on what value you choose, includes most average size PDCs and other proximal hazards
- For many volcanoes, this is enough



Energy cone (or ΔH/L cone) for PDCs

- Trying to find/predict the runout of PDCs of varying sizes
- ΔH/L value used has to do with the volume of the PDC
- Lower values for larger PDCs (0.1-0.2) and higher values for smaller PDCs (0.2-0.5)
- Important if you want to show energy cones for different size PDCs
- For volcanoes with large PDCs, or populations that live very close to the volcano, it might be useful to have more than the 1 proximal hazard zone, and instead add another 1-2 energy cones representing larger (or smaller) volume PDCs





Within 5 km	0
Within 10 km	128
Within 30 km	3,187

Within 5 km	997
Within 10 km	13,801
Within 30 km	163,630

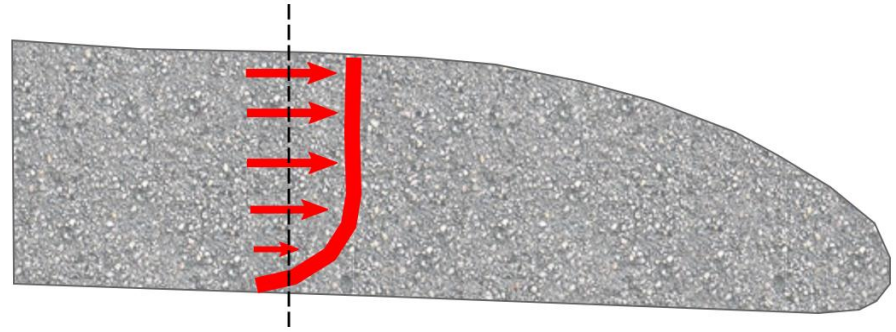
Possible Lahar and PDC Modeling Workflow

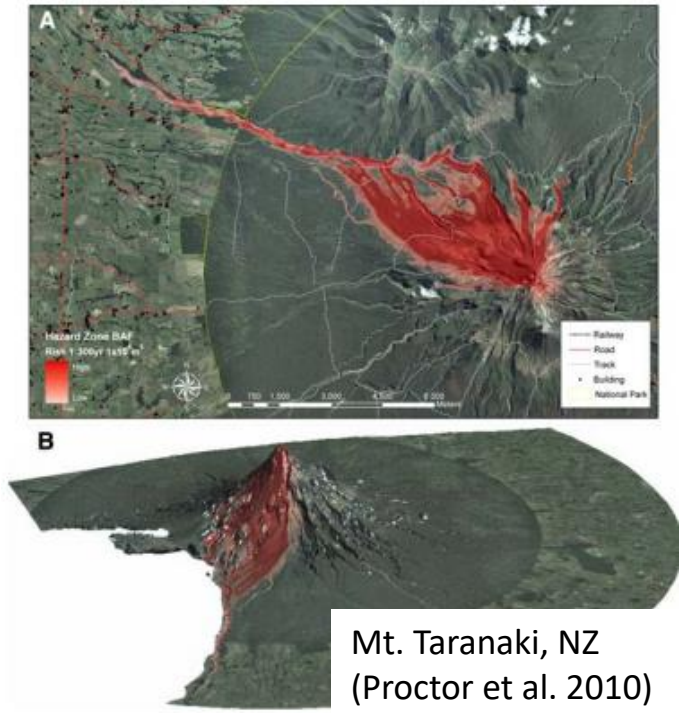
For volcanoes with moderate PDCs/low population density:
1. Run LAHARZ using an H/L cone that identifies where lahars will switch from deposition to erosion. This is sort of like identifying where the break in slope is at your volcano. It is usually between 0.1-0.3 and will depend a bit on the size of the volcano. Use the starting points it finds to run LAHARZ for your desired range of lahar volumes (you can always edit the starting point list down to one point per stream, it usually identifies too many points). Display this H/L cone on the map and call it the 'proximal hazard zone' for PDCs and ballistics. Done!

For volcanoes with large PDCs, or high population densities near the volcano:
2. Create more PDC hazard zones instead of just one proximal hazard zone. Use the H/L cone from step 1 to run LAHARZ. Choose some PDC volumes, maybe 0.001, 0.01, 0.1 km^3 (depends on your volcano). Using these volumes, use the regression equations to find what an appropriate H/L value is for these PDC volumes. Use the LAHARZ proximal hazard zone tool to make H/L cones for these values (but don't run LAHARZ again from these points!). Display these different H/L cones on the map and label them something like 'High probability PDC zone' for the smallest volume/smallest H/L cone, etc.

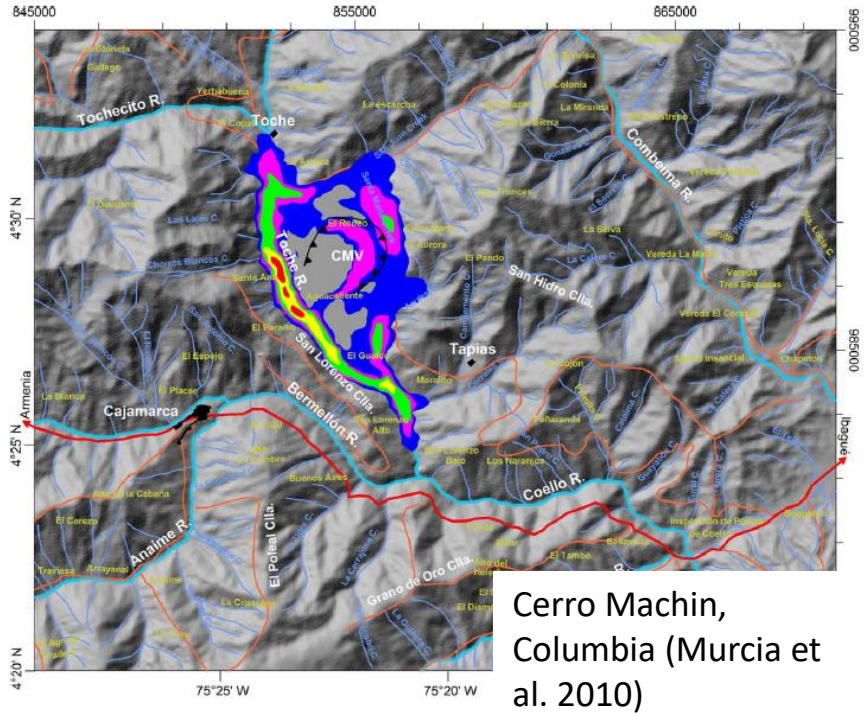
TITAN2D

- TITAN2D is a computational model that simulates dry granular mass-flows over DEMs of natural terrain.
- No ash-cloud surge!
- Depth-averaged equations for the conservation of mass and momentum use Mohr-Coulomb friction terms.
- Volume, basal and internal friction are model inputs
- Comparisons of TITAN2D with both experimental and real pyroclastic flows have shown success in a wide variety of settings (Pitman et al. 2003; Patra et al. 2005; Rupp et al. 2006, etc.).





Mt. Taranaki, NZ (Proctor et al. 2010)



Cerro Machin, Columbia (Murcia et al. 2010)

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TITAN2D is very sensitive to a number of input parameters:

- Volume (volume of the total collapse)
- Basal friction angle
- Initial starting coordinates
- Initial direction
- Initial pile geometry

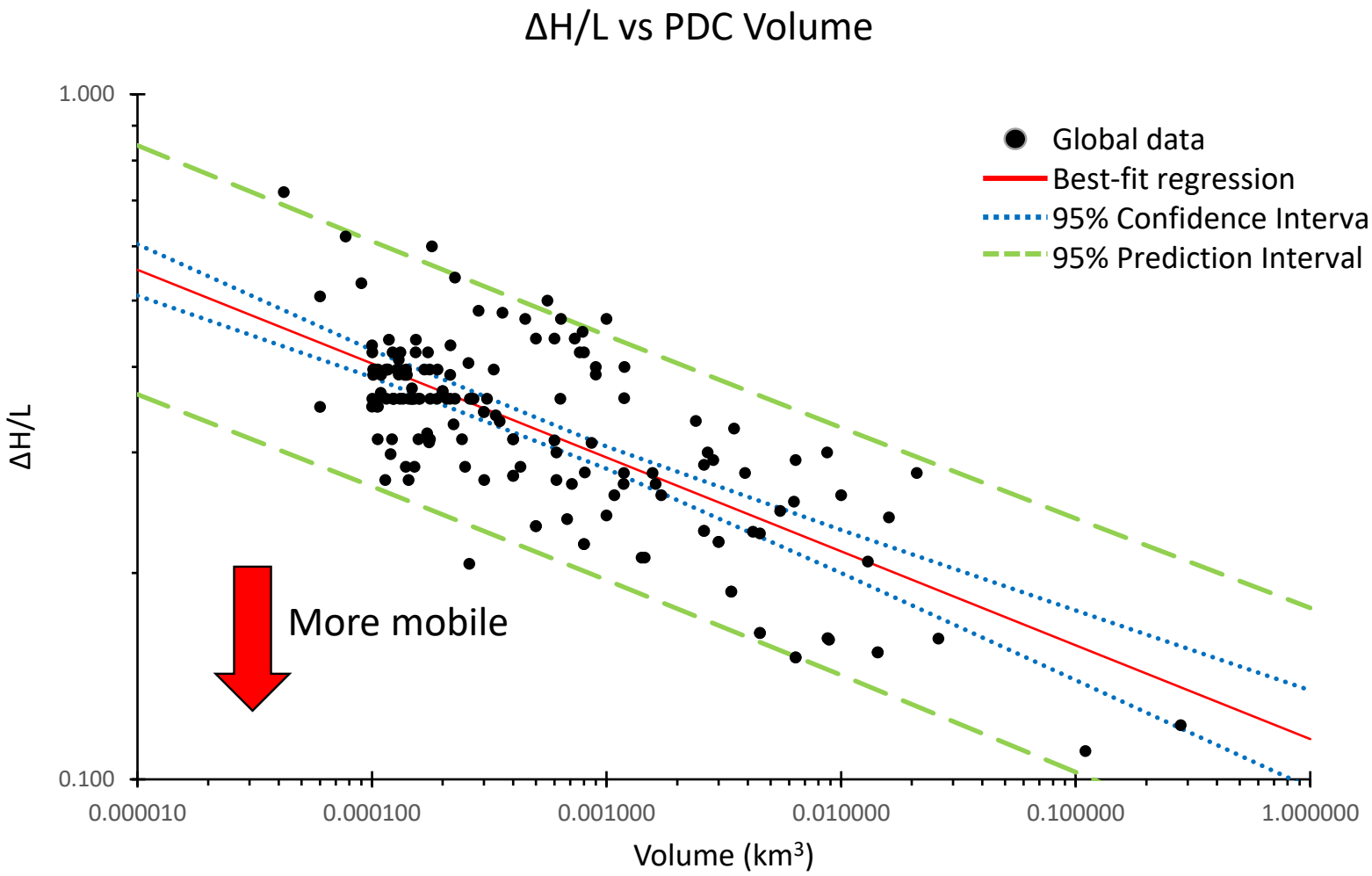
Because of this, it is **extremely important to vary these parameters for different model runs.**

Basal friction

The ΔH/L values for PDCs of different volumes work well for TITAN2D basal friction parameters

Convert ΔH/L values to degrees by:

Degrees(atan(ΔH/L value))



Best fit: $\Delta H/L = 0.10(\text{Volume}^{-0.15})$

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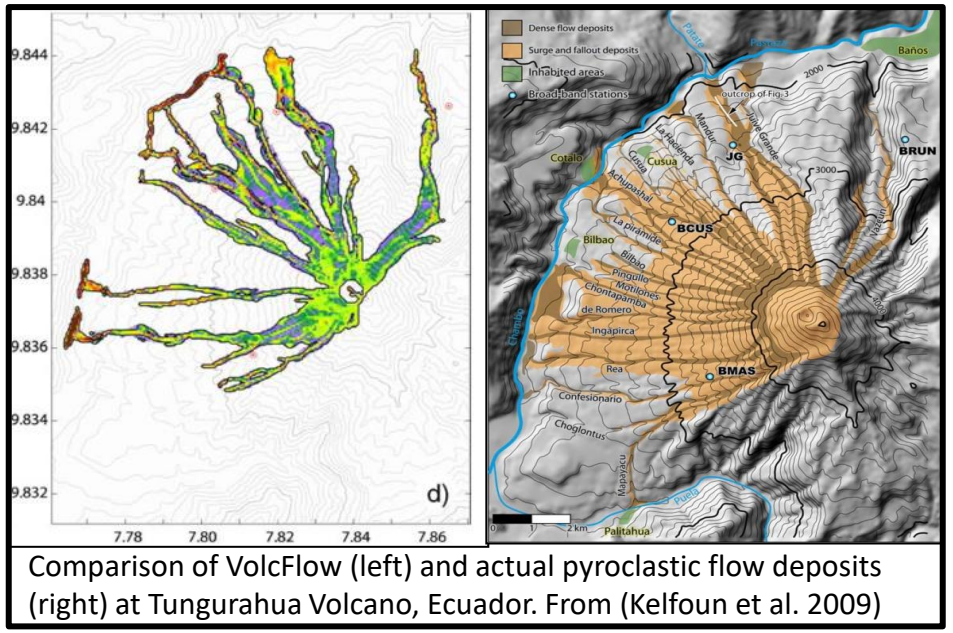
TITAN2D basal friction parameters from FlowDat regressions

Volume (km^3)	Volume (10^6 x m^3)	Dome-collapse friction (deg)	Collumn-collapse friction (deg)
0.00001	0.01	28 (27-31)	34 (27-42)
0.0001	0.1	21 (20-22)	25 (21-29)
0.001	1	16 (15-16)	23 (17-31)
0.01	10	12 (11-12)	15 (12-19)
0.1	100	8 (7-9)	10 (8-12)

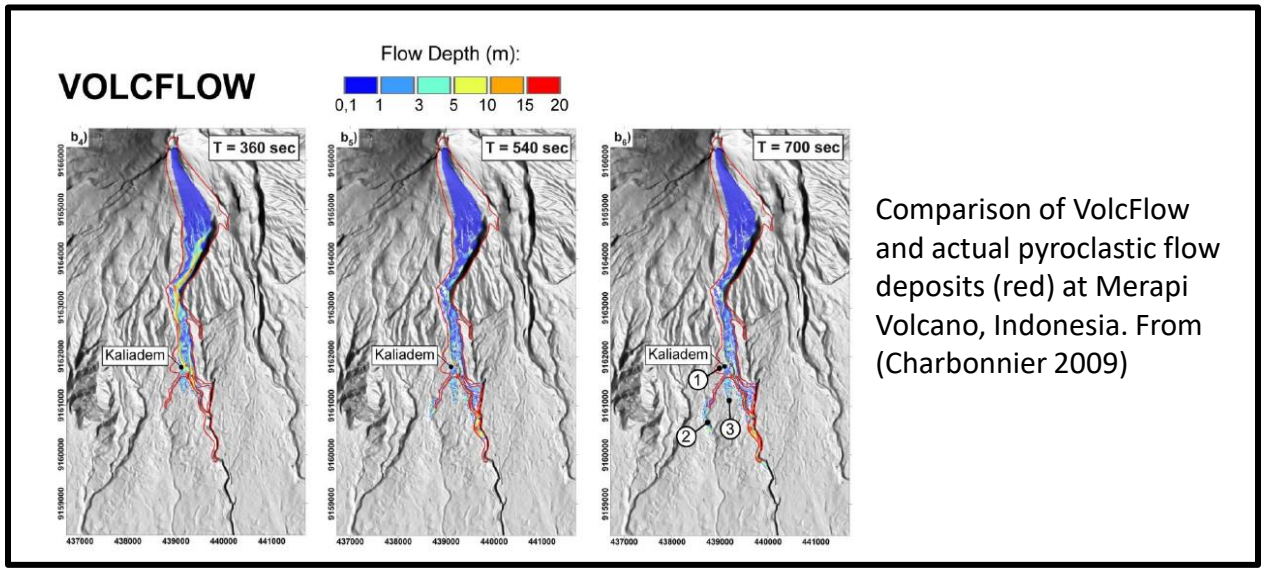
Ranges in parentheses reflect the uncertainty
For TITAN2D runs, I usually use the lower end of the range!

VolcFlow

- VolcFlow solves depth-averaged equations of mass and momentum using a DEM.
- No ash-cloud surge in basic version!
- One stress parameter accounts for all retarding stresses
- This stress parameter can be held constant for the constant retarding stress condition, so that the ratio of driving to retarding stress decreases as thickness increases
- Takes volume, CRS, and bulk flow density
- Volcflow has shown success in replicating pyroclastic flows at Tungurahua (Kelfoun 2009) and Merapi volcano (Charbonnier 2009).



Comparison of VolcFlow (left) and actual pyroclastic flow deposits (right) at Tungurahua Volcano, Ecuador. From (Kelfoun et al. 2009)



Comparison of VolcFlow and actual pyroclastic flow deposits (red) at Merapi Volcano, Indonesia. From (Charbonnier 2009)

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All these models are averaging a number of variables that are actually highly unsteady and non-uniform in nature, especially in sustained, larger-volume flows with significant ash-cloud surge components.

Models cannot capture the effects of:

- increasing excavation of pressurized dome material
- interactions between subsequent flow pulses
- evolving pore-fluid pressures and granular temperatures
- variations in frictional vs. collisional regimes
- sedimentation, erosion, or deposition
- polydispersity of grain sizes, shapes, and densities
- **interplay between dense-basal undercurrent and ash-cloud surge**

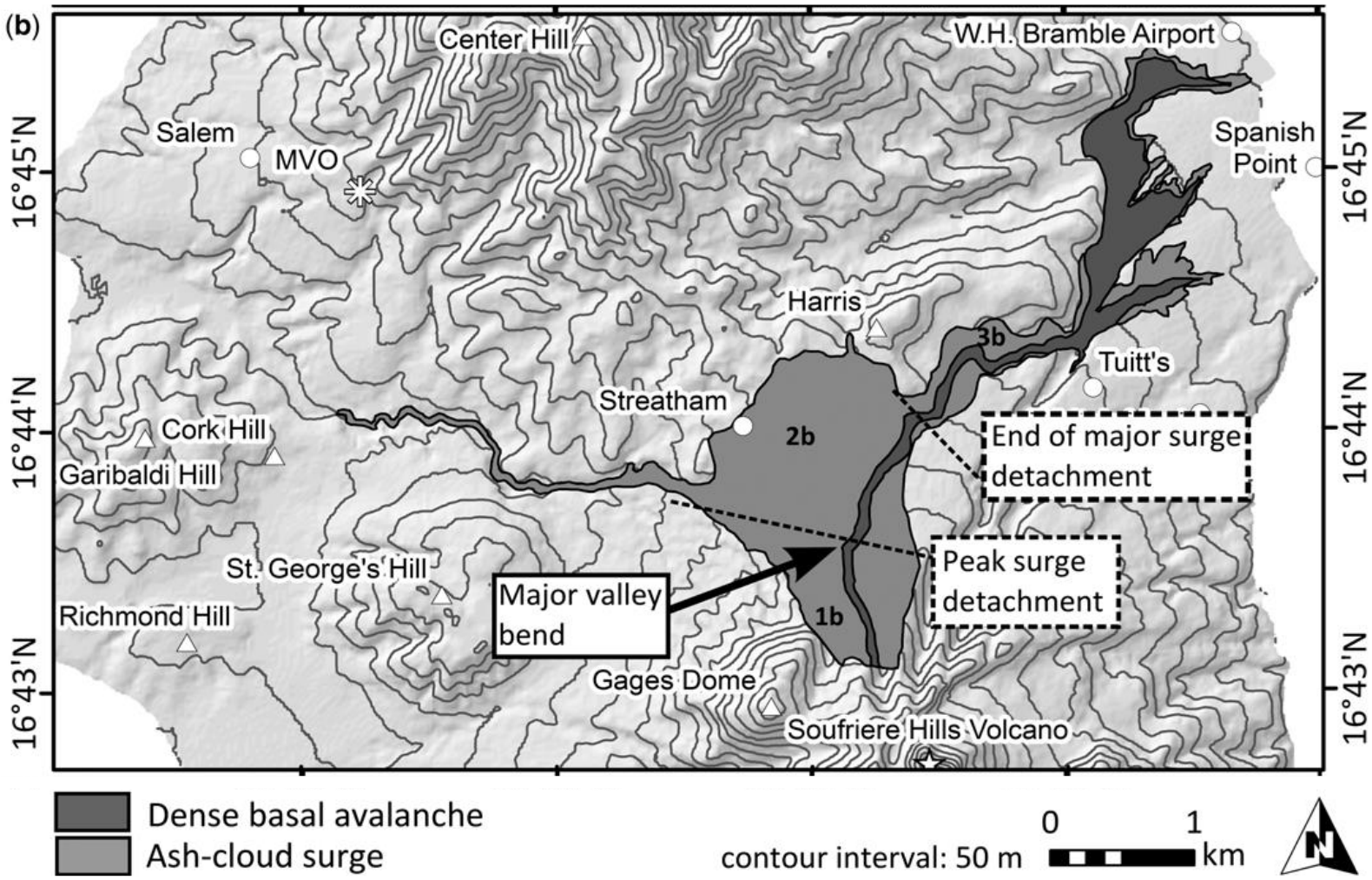
PDCs probably do not exhibit a fixed rheology, but evolve as fluid-pressure, grain concentration, and granular temperature change through time and space

These effects are probably most pronounced in large-volume PDCs, especially for column-collapse flows

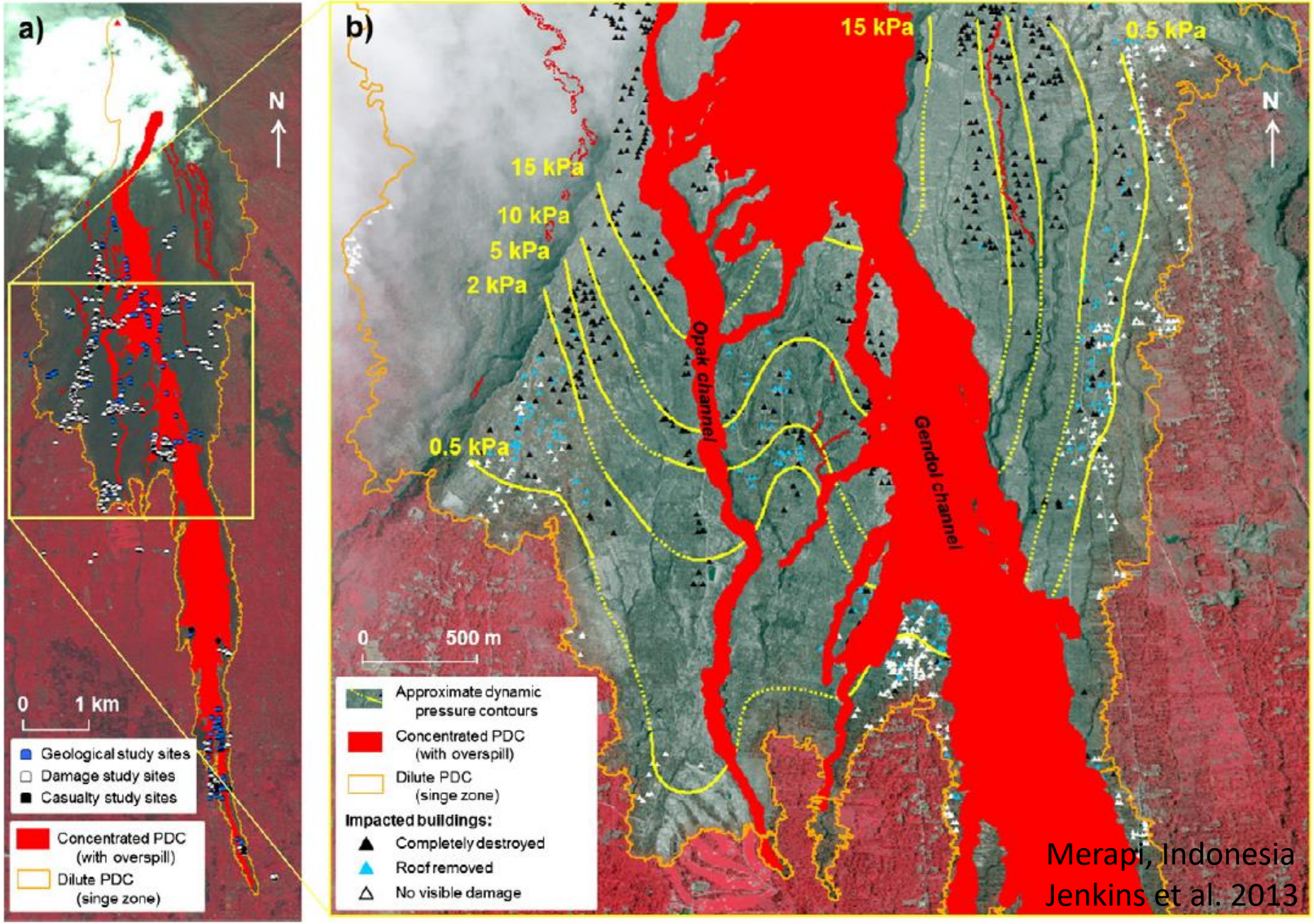
Some multiphase models can model these things, but they are research tools, not hazard simulation tools because they take so long!

Ash-cloud surge

- Can detach from the dense basal avalanche at constrictions in channels or valley bends and travel long distances over relatively flat ground
- Decoupled surges often responsible for deaths (Soufrière Hills Volcano, Merapi, Unzen, Sinabung, Pelée)



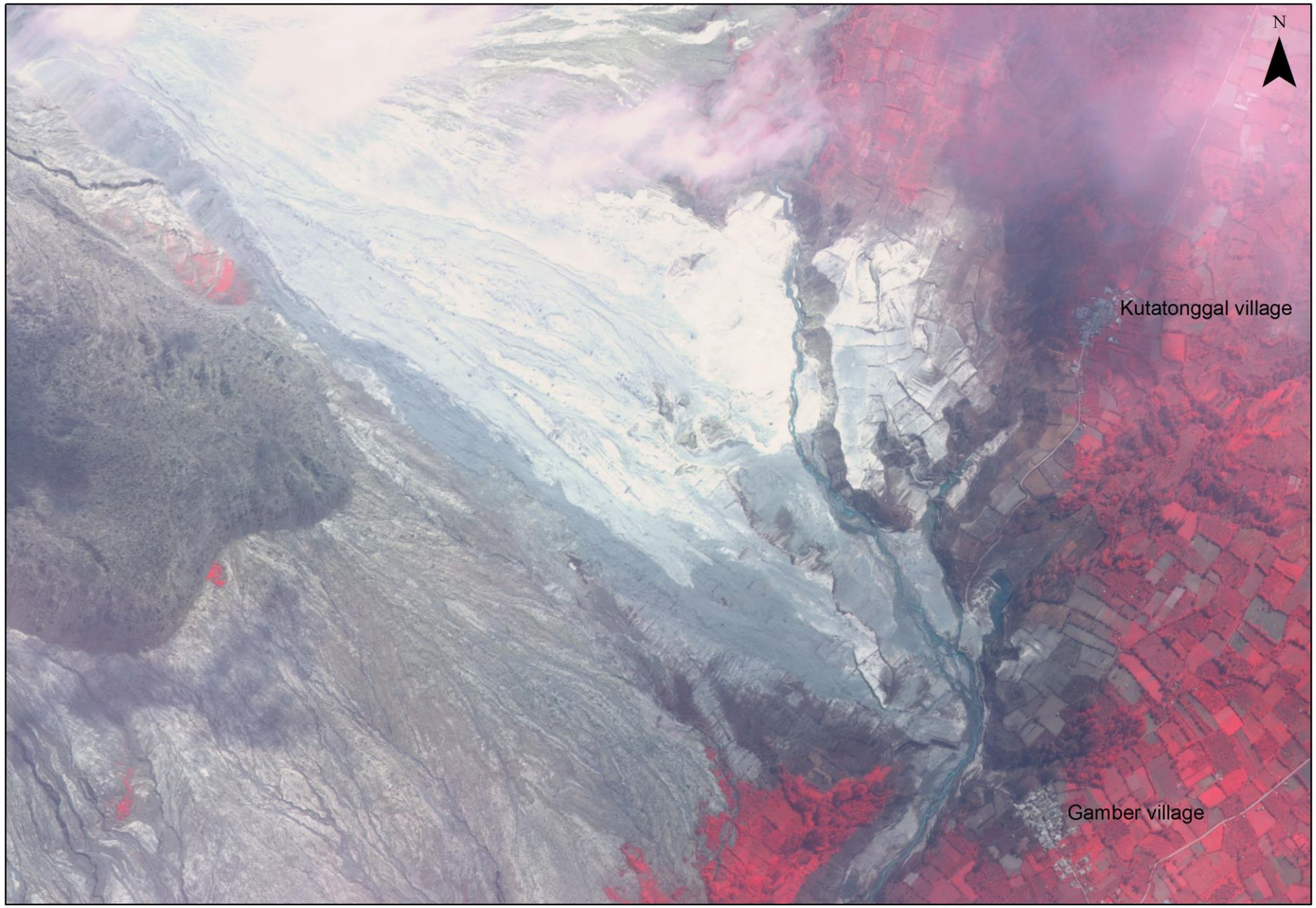
Soufrière Hills Volcano, Montserrat
Ogburn et al. 2014



Merapi, Indonesia
Jenkins et al. 2013

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Sinabung Volcano 12 May 2016 WorldView 3

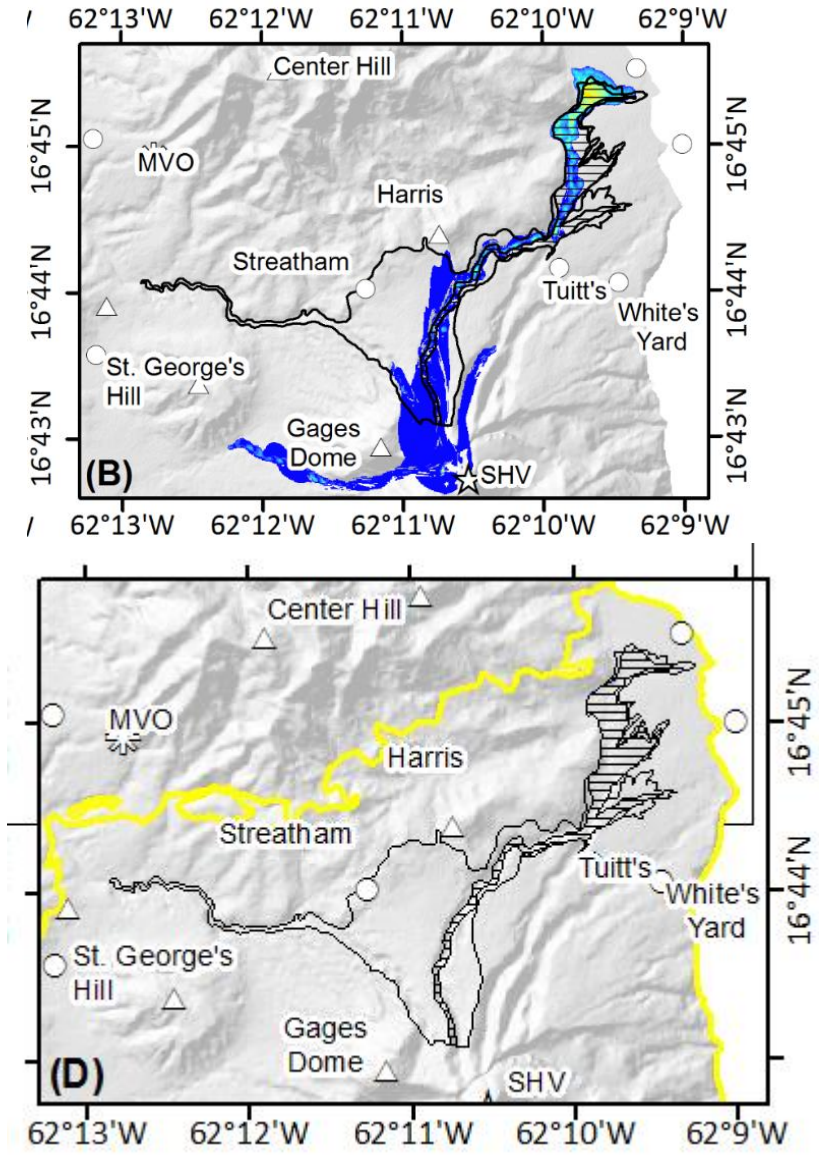


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0 0.1 0.2 0.4 0.6 0.8
Kilometers

Ash-cloud surge

- TITAN2D DOES NOT model the overriding, dilute, ash cloud surge component
- ΔH/L cones don't explicitly model the surge, but they **may** encompass surge hazards because they don't simulate a channelized flow



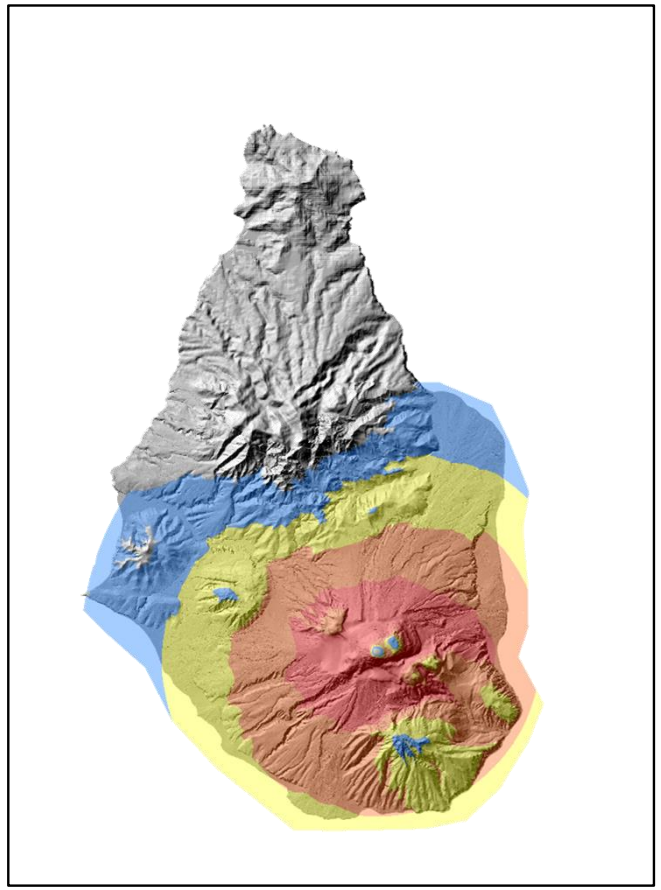
Model	Pros	Cons
Energy cone	• With local ΔH/L vs. volume data, does a decent job reproducing flow runout • Often captures ash-cloud surge hazard area too, because it's not modeling channelization • Extremely fast (< 1 minute) • Very small output file sizes • Very well-documented • Only need to run Energy Cones for a range of volumes to produce a hazard map	• Reliant on ΔH/L vs. volume data, if data is sparse for a particular volcano, relies on widely scattered global data which results in large spread in runout distances • Might be overly-conservative because there's no channelization
TITAN2D	• Input parameters obtainable from deposits, observations • Reasonable reproduction of inundation and velocity • Reproduces gradually thinning and spreading margins of large volume deposits • Small file size • Can be run using only open-source software, either online at Vhub.org or on a linux machine • Very well-documented	• Requires accurate summit/dome DEM • Sensitive to pile shape, geometry • High velocities make things leave valleys when they shouldn't • No ash-cloud surge modeling • Slow (4-8 hours) • Requires many model runs to produce hazard map
VolcFlow	• Reproduces inundation, thickness, and velocity very well • Reproduces well-defined margins of small volume flows • Faster than TITAN2D (1-2 hours)	• CRS related to thickness, but poorly understood • Doesn't reproduce gradually thinning and spreading margins of large volume deposits very well • No ash-cloud surge modeling • Poorly documented • Extremely large total file size (up to 100GB) • Requires MATLAB, Surfer • Requires many model runs to produce hazard map

Using models for hazard mapping

ONE MODEL RUN ≠ HAZARD MAP

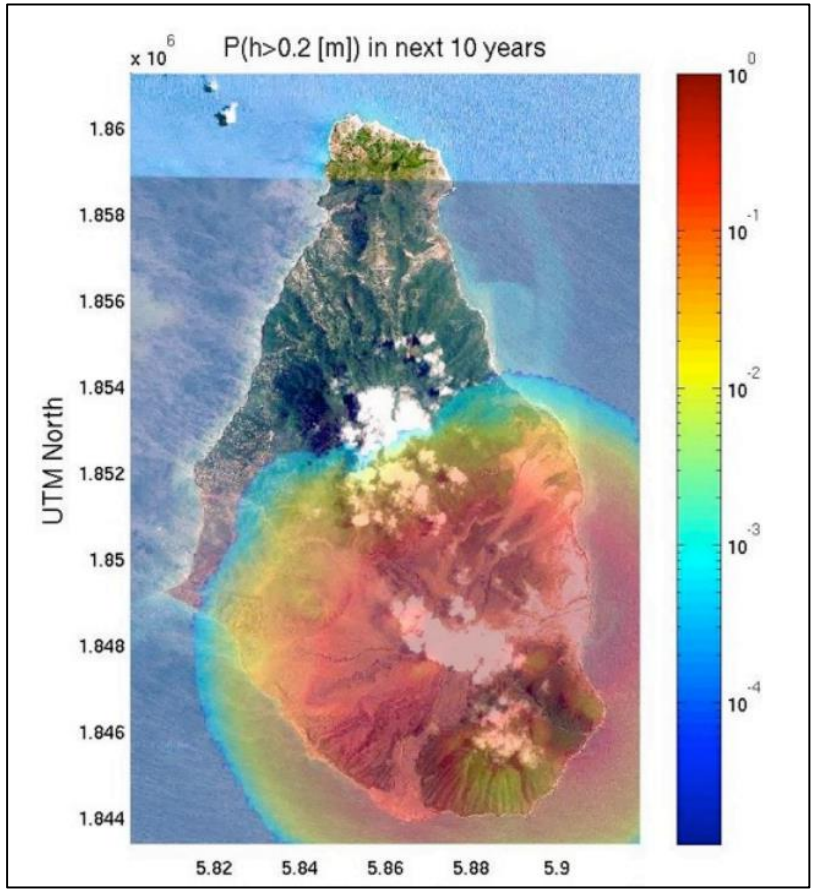
- For hazard mapping, the input parameters ***must*** be varied
- **For LAHARZ:** use a variety of volumes down different valleys
- **For energy cone:** use a variety of ΔH/L values to show hazard from different PDC volumes
- **For TITAN2D/VolcFlow/other physical models:** use a variety of volumes, basal frictions angles, starting locations, and directions (fully probabilistic hazard maps require 1000's of runs using a sampling of input parameter distributions)
- Compare multiple models and consider geologic evidence of past flows
- There's lots of uncertainty in modeling PDCs, especially column-collapse flows!

Energy Cone



4 ΔH/L energy cones
(for 0.0001 to 0.1 km^3)
< 5 minutes on 1 processer

TITAN2D



100,000 TITAN2D model runs
(for 0.0001 to 0.1 km^3)
~1000 hours on 1024 processers

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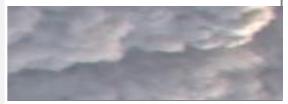
Questions?

	Energy Cone	TITAN2D/VolcFlow
DEM quality	DEM is old or poor resolution	DEM is new & good resolution
How much time do you have?	Limited time	Lots of time
What is your primary goal?	Hazard mapping	Understanding physics, answering a science question, exploring PDC behavior in particular valleys, or probabilistic hazard mapping (with lots of time for many runs)
Ash cloud surges	People live close to valleys, in interfluves, ash-cloud surge overbanking is a concern	You're only concerned about runout and not ash-cloud surge spillage out of valleys
Expertise and resources	You're a smart beginner with any computer	You're an expert with lots of computing resources

How to use TITAN2D (or other geophysical models) when you can't do thousands of model runs

- Compare TITAN2D runs with similar ΔH/L cones for the same volumes
- Use TITAN2D to see if only certain sectors will be affected (be sure to try different initial locations, directions, velocities to be sure)
- Use TITAN2D to see if there are topographic barriers that the energy cone can't 'see'
- Use TITAN2D to ask specific questions:
  - How big of a flow would it take to reach point X
  - What parameters would it take to make a flow go to the west

Model	Inputs	Rheological model	Assumptions	Solves	Outputs
Energy cone	DEM Starting point (can choose max elevation) **ΔH/L (derived from volume and global stats)**	Sliding block model, ΔH/L inversely proportional to V	Flows are well described by sliding block model, empirical relationship applies to all volcanoes	Draws an energy cone from user-supplied max elevation to elevation described by ΔH/L	Runout, as an energy cone
TITAN2D	DEM Volume Initial geometry and velocity Vol flux if running as a flux source **Friction (basal and internal)**	Mohr-Coulomb friction	Depth averaged	Equations of mass and momentum	Thickness Velocity Inundation area (can calculate KE)
VolcFlow	DEM Volume Initial geometry and velocity **Constant retarding stress** **Density**	Constant retarding stress	Depth averaged	Equations of mass and momentum	Thickness Velocity Inundation area (can calculate KE)



Column-collapse modeling

- Difficult to know how much of the column will collapse vs. stay aloft/become ashfall
- So there is a lot of uncertainty between VEI and column-collapse flow volume
- Also, because the column-collapse flows spread out into all channels radially, it's hard to know how much volume will go down each valley, and how far